

CC1/CC2 direct to CH224K (internal Rd). DP/DM of CH224K shorted together and NOT connected to the connector (WCH 'PD-only' mode) so USB D+/D- go to the MCU. CFG2/CFG3 float = single-resistor mode. CFG1 56k = 15V. PG low = contract OK. CH224K VBUS sense pin left NC (allowed in PD-only mode; pin is only 13.5V rated).

3V3 = OR of: VBUS (charger OR a phone in the USB-C port) | battery via Q2 when the MCU asserts PWR_HOLD (lets the MCU finish balancing, put BQ25798 in shutdown + BQ76920 in SHIP after unplug, then release itself). Nothing else can power the MCU from the pack. Idle battery drain ~1-2uA total.

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PROG 10.5k -> 3S / 1.5MHz; POR defaults VREG=12.6V, ICHG=1A, VYSMIN=9V.
NO ACFTETS: VAC1/VAC2 tied to VBUS, ADCRV1/2 to GND, ILIM_HIZ->RGEN (limit set by i2C)
/CE held HIGH by 100k -> NO charging until MCU drives CHG_EN high (after cell check).
BQ25798 D+/D- left open: MCU must clear AUTO_INDET_EN and set INDPMP over i2C.
TS: 5.23k/30.1k + 10k NTC = 0.60C window (datasheet 7.3.9.5). Shutdown mode = 0.5uA.

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PF0/PF1 = I2C1 SDA/SCL (AF1) PA11/PA12 remapped onto pins 17/18 for USB (SYSCFG PA11-PA12_RMP).
PA1/PA3/PA0 = charlieplexed cell LEDs A/B/C (pads TP13-15). Charger /INT, CH224K PG and BMS ALRP are not
wired to the MCU (firmware pins I2C). PA2 CHG_EN, PA4 BMS boot, PA5 VBUS/11 ADC, PA6 LED, PA7 PWR_HOLD.
PB1 = FAST_CHG_EN (JP2 to GND, internal pull-up): open = slow charge (default), bridged = fast charge.
Bridge JP1 + plug USB = ST ROM DFU bootloader. Backup: ST-Link on test pads TP5 SWDIO, TP6 SWCLK, TP7 3V3, TP8 GND.

3S per dataset Table 9-2: VC5-VC4 = cell3, VC4-VC3=VC2 shorted, VC2-VC1 = cell2, VC1-VC0 = cell1.
 Cc=56R / Cc=1uF cell filters -> ~37mA internal balancing (dataset min Rc 40R, max 50mA) (never balance adjacent cells together).
 SRP/SRN to VSS (no coulomb counter), CHG/DSG unused, Sits in SHIP mode (~0.6uA) when idle.
 Balance lead also carries the charge current (pins 1 and 4): firmware pauses charging before each cell reading.